ST FRANCIS CATHOLIC SECONDARY SCHOOL, LAGOS

The Jesuit-Run Catholic College

DIGITAL TECHNOLOGIES

JSS 1

First Term

2026/2027 session

SCHEME OF WORK

WEEKS	TOPICS
1	Historical Development of Digital Devices
2	Career opportunities in IT
3	Basic Computer Hardware Components I
4	Basic Computer Hardware Components II: Storage Units
5	MS Word
6	Digital Media
7	Safety and Security Features on Digital Devices
8	Continuous Assessment / mid-term break
9	Management of e-Waste
10	Introduction to Graphic Design and Canva

Date	**WEEK 1**
Topic	Historical Development of Digital Devices
Objectives	• Define a computer and digital device. • Identify the parts of a computer. • Explain the stages in the historical development of digital devices. • Describe the generations of computers and their characteristics.
Insight	Digital devices evolved from simple counting methods such as fingers, pebbles, and cowries to sophisticated electronic computers powered by artificial intelligence. Each stage of development improved speed, accuracy, storage capacity, and efficiency, leading to modern computers and smart devices used today.
Key words/New terms	Computer, Digital Device, Binary Code, Program, System Unit, Peripherals, Punch Cards, Supercomputer, Vacuum Tubes, Transistors, Integrated Circuits (ICs), Microprocessor, Artificial Intelligence (AI), Natural Language Processing (NLP), Internet of Things (IoT).
Application	• **Improving innovation:** Learning from past inventions inspires scientists, engineers, and programmers to develop better technologies. • **Solving problems efficiently:** Knowledge of earlier computing methods helps us understand and improve current computer systems. • **Making informed technology choices:** Understanding different generations of computers helps users select appropriate devices for their needs.
Summary	A computer is an electronic device that processes data according to instructions called programs. • A computer consists of the system unit and peripherals. • Digital devices use binary code (0s and 1s) to process and store information. • The history of digital devices progressed through Early Counting Devices, Mechanical Devices, Electro-Mechanical Devices, and Electronic Devices. • Important inventors include Blaise Pascal, Gottfried Leibniz, Joseph Jacquard, Charles Babbage, Herman Hollerith, John von Neumann, and Philip Emeagwali. • Computers have evolved through five generations: Vacuum Tubes, Transistors, Integrated Circuits, Microprocessors, and Artificial Intelligence. • Modern computers are faster, smaller, more reliable, and capable of intelligent tasks such as speech recognition and natural language processing.
Student engagement: *i.* What have I learnt? *ii*. Skills acquired	
My Question(s)/Evaluation .	
Action	

HISTORICAL DEVELOPMENTMENT OF DIGITAL DEVICES

A computer is an electronic device that can accept data and process the data according to some stated instruction and generates results as information. The instructions that are given to the computer to process data are called programs.

PARTS OF A COMPUTER

The typical computer system consists of several physical parts which can be grouped into two, namely:

1. THE SYSTEM UNIT: This is the main enclosure that contains the computer's primary electronic components. These components include Motherboard, CPU (Central Processing Unit, RAM (Random Access Memory), Storage Drives (hard disk drive or solid-state drive), Power Supply Unit (PSU) etc.

2. PERIPHERALS: Peripherals are supplementary hardware components that enhance the computer's functionality and allow you to interact with the system unit keyboards, mice, webcams, and scanners, monitors, printers, and speakers, USB flash drives or external hard drives

HISTORICAL DEVELOPMENTMENT OF DIGITAL DEVICES

Digital device is a term used to describe computer and electronic-based equipment that transfers information by breaking it down into numerical code. The numbers, or "digits," in digital technology are based on a system called binary code. This code uses a combination of 1s and 0s to record information such as sounds, words, and images. The information can then be sent from one source to another and reassembled on the receiving end. The process allows larger amounts of information to be condensed, stored, and transmitted quicker and more efficiently. Advances in digital capabilities have led to revolutionary new technology such as the internet and smartphones and changed the daily lives of people throughout the world.

Stages in the History of digital devices

The evolution of the computer is generally divided into several key stages, each marked by significant advancements in computational methods and technology:

1. Early Counting Devices
2. Mechanical Counting and calculating Devices
3. Electro-mechanical Counting Devices
4. Electronic Counting Devices

EARLY COUNTING DEVICES

Long ago, people used various simple, readily available, and affordable counting methods to solve their daily counting needs. These included natural elements and simple tools. Some of these early counting devices are examined below:

1. Fingers and Toes
2. Pebbles
3. Grains
4. Cowries
5. Canes
6. Writing on the Wall

Major Problems Associated with the Use of Early Counting Devices

While these early methods were useful, they had several significant limitations:

- They could not be used efficiently for counting very large numbers.
- They often required a lot of physical space for storage (e.g., piles of stones).
- Their use was time-consuming for complex calculations.
- They were not always readily available in all environments.
- They were prone to human error, making them not always accurate.
- They were not very versatile, meaning they could only perform a limited number of basic calculations.

MECHANICAL COUNTING AND CALCULATING DEVICES

- **Abacus:** An abacus (or counting frame) is a traditional, hand-operated calculating tool used to perform mathematical operations such as addition, subtraction, multiplication, and division. It consists of a rectangular frame with parallel rods and slidable beads.

- **Slide rule**: A slide rule is a mechanical calculating device that uses logarithms to perform mathematical operations like multiplication, division, exponentiation, and more. It was invented by William Oughtred in England in 1620. It consists of two main parts: a fixed body with logarithmic scales and a movable slide with corresponding scales.

- **Napier bone:** This was an invention of John Napier (a Scottish mathematician) in 1617 for performing arithmetic operations like multiplication and division of large numbers.

ELECTRO-MECHANICAL COUNTING DEVICE

These are counting and calculating devices that combine the use of electricity and machines to operate.

- **Pascal calculating machine**: This was a calculating machine used for addition and subtraction only. It was called pascaline and was invented in 1642 by a French mathematician and philosopher called Blaise Pascal.

- **Leibniz Machine (Stepped Reckoner):** In 1671, German mathematician Gottfried Leibniz built a more advanced machine that made it easier and faster to perform basic arithmetic operations. He named it the Leibniz Stepped Reckoner. The Leibniz calculator was the first true calculator that performs the functions of addition, subtraction, multiplication, and division. The process of multiplication involved repeated addition and division involved repeated subtraction. The Leibniz calculating machine was an improved version of Pascaline.

- **Jacquard loom:** The Jacquard Loom was invented by Joseph Marie Jacquard in 1801. This revolutionary machine used punched cards to control the weaving of intricate patterns into fabric. Although the loom itself was not used for mathematical computation, its use of punched cards for programmed control is considered a highly significant step in the history of computing, paving the way for later programmable machines.

- **Charles Babbage difference engine and analytical engine:**
  The analytical engine is a general-purpose programmable mechanical device proposed and designed by Charles Babbage. Charles Babbage was an English mathematician and a professor at Trinity College, Cambridge, England. He conceived and proposed two mechanical machines:

  1. The difference engine
  2. The analytical engine

  The **difference engine** was a mechanical calculator that tabulates polynomial functions using Newton's method of divided differences. However, in 1833, Babbage could not continue the development of the machine due to a lack of funds. The **analytical engine** was conceived in 1834 and was proposed to be a programmable multipurpose calculating device. With the concept of Jacquard's loom, he designed a machine that could solve complex problems through controlled programs.

  The analytical engine was designed to have four main parts which are used by the modern-day computer:

- The punch cards or readers were used to enter data into the machine. This becomes the keyboard in a modern computer.
- The mill which performs the computations. This essentially translates to the processor.
- The store where numbers and computation results are kept. This translates to computer memory.
- The output or printer which gets out the results. The output could be in hardcopy printouts, punched cards, and stereotypes.

Unfortunately, Babbage could not completely build his proposed machine until he died in 1871. However, his idea and concept gave birth to the modern computer. He is therefore regarded as the **"father of the computer".**

ELECTRONIC COUNTING DEVICES AND MODERN COMPUTERS

These are devices that are purely electronic in operation. They include;

- **HERMAN HOLLERITH'S PUNCH CARDS**: Herman Hollerith's punch card device is an electronic tabulation machine used in statistical calculation and analysis. It was invented by Herman Hollerith in 1890 and was used for USA census in 1890.

- **The Von Neumann Machine:** John von Neumann was a brilliant mathematician and computer scientist who came up with a very important idea: **the stored-program computer** in 1949**.** This meant that both the instructions for the computer and the data it was working on could be stored in the computer's memory. This made computers much more flexible and powerful. Almost all computers today follow von Neumann's basic design. While not a specific device itself, it was an architectural concept that enabled the development of modern electronic counting devices.

The Evolution of Modern Computers

Building on the principles of earlier counting devices, scientists began to create the first true computers. These were massive machines by today's standards, but they represented a giant leap forward in computing power.

- **ENIAC (Electronic Numerical Integrator and Computer):** Often considered the first general-purpose electronic digital computer, ENIAC was huge, occupying a large room and weighing tons. It used vacuum tubes and was a major advancement, although it still used decimal numbers (not binary).

- **Manchester Mark 1:** This computer, built in England, was one of the first to use the stored-program concept pioneered by von Neumann. This was a crucial step toward the architecture of modern computers.

- **EDSAC (Electronic Delay Storage Automatic Calculator):** Inspired by von Neumann's ideas, EDSAC was another early stored-program computer. It further refined the design and demonstrated the power of this approach.

- **EDVAC (Electronic Discrete Variable Automatic Computer):** A successor to ENIAC, EDVAC was also developed with the stored-program concept in mind. These machines were getting closer and closer to the computers we know today.
- **UNIVAC (Universal Automatic Computer):** UNIVAC was one of the first computers built for commercial use. It marked the beginning of computers moving out of the lab and into businesses and other organizations.

- **Supercomputers and Philip Emeagwali**: As technology continued to advance, computers became faster and more powerful. This led to the development of supercomputers, which are designed to perform extremely complex calculations.

  Philip Emeagwali: A Nigerian-American computer scientist, Emeagwali is known for his pioneering work in supercomputing. He developed innovative ways to connect thousands of microprocessors to work together, creating incredibly fast computers. His work has had a significant impact on fields like oil exploration and climate modelling.

GENERATION OF COMPUTERS

The five generations of computers represent a massive evolution from room-sized, power-hungry machines to smart, portable devices. They are defined by their core processing technologies:

First Generation (1940–1956): Vacuum Tubes

Core Technology: Used vacuum tubes as the internal circuitry and magnetic drums for memory.

Size & Heat: Massive, often occupying entire rooms, and generated extreme amounts of heat requiring heavy air conditioning.

Speed & Language: Extremely slow (measured in milliseconds) and relied on difficult-to-use machine language (binary 0s and 1s).

Input/Output: Relied heavily on punched cards and paper tape.

High Power Consumption: Vacuum tubes required a high amount of power or electricity to operate. This contributes to high energy costs and heat output.

Expensive: The cost is a major problem in this generation. The cost of vacuum tubes and the complexity of building these computers made them expensive. That's why these computers are accessible only to large organizations in big areas.

Examples: ENIAC, IBM 701, Manchester Mark 1, EDSAC, EDVAC, UNIVAC-I and II.

Second Generation (1956–1963): Transistors

Core Technology: Replaced bulky vacuum tubes with smaller, faster, and more reliable transistors.

Efficiency: Much smaller, faster, more energy-efficient, and generated less heat.

Size and Cost: Computers in second-generation are smaller in size and less costly than first-generation. However, still everyone can't afford it.
Language: Transitioned from machine language to assembly language and early high-level languages like COBOL and FORTRAN.

Usage: Shifted from purely scientific or military use toward basic business and commercial applications.

Examples: IBM 1401, Honeywell 400, IBM 7094, UNIVAC 1108 and CDC 1604

Third Generation (1964–1971): Integrated Circuits (ICs)

Core Technology: Used Integrated Circuits (ICs), where thousands of transistors were etched onto a single silicon chip.

Speed & Size: Drastic leap in processing speed (measured in microseconds) and became small enough to fit on standard desks.

High-Level Languages: Due to complex programming previous generations had a lot of difficulties. Third generation is easier with the adoption of high-level languages. Languages such as FORTRAN, BASIC, and COBOL. These languages help developers to maintain more human-readable code.

Interface: Eliminated punch cards in favour of keyboards and monitors, and introduced operating systems that could run multiple tasks (time-sharing).

Reduced Power Consumption and Heat: Reduced Power Consumption and Heat is only happening due to IC technology. And the results are impressive, making computers more energy-efficient.

Examples: IBM 360, IBM 370, PDP-11, UNIVAC 1108**.**

Fourth Generation (1971–Present): Microprocessors

Core Technology: Built around microprocessors, placing the entire Central Processing Unit (CPU) on a single, tiny chip (LSI and VLSI technology).

Accessibility: Spawned the Personal Computer (PC) revolution, making computers highly affordable for individuals and small businesses.

Software & UI: Introduced Graphical User Interfaces (GUIs), allowing users to click icons and use a mouse instead of typing text commands.

Networking: Enabled the rise of the internet, LANs, and portable devices like laptops and smartphones.

Fifth Generation (Present and Beyond): Artificial Intelligence

Core Technology: Powered by **Artificial Intelligence (AI).** Fifth generation computers are good at mimicking human intelligence. This includes speech recognition, image recognition, and natural language processing.

Processing: Utilizes **parallel processing** (multiple microprocessors working simultaneously) to process massive amounts of data in real-time.

Interaction: Features **Natural Language Processing (NLP)**, meaning computers can understand and respond to human speech and gestures (e.g., Siri, Alexa).

Connectivity: Deeply integrated with **cloud computing**, the Internet of Things (IoT), and highly advanced graphics processing.

Examples: **Watson, Google DeepMind (AlphaGo), Siri, Alexa, Google Assistant, ChatGPT**

Date	**WEEK 2**
Topic	**Career opportunities in IT**
Objectives	By the end of the lesson, students should be able to: i. Revise and discuss the first terms learning ii. Partake in welcome back test. iii. Identify career opportunities in IT; iv. State the qualities of an IT Professional v. List different computer Professional bodies
Insight	Computer-related careers offer diverse and growing opportunities across software, data, cyber-Security, AI, and IT in nearly every industry.
Key words/New terms	Career, Professional, Professional Bodies**,** Technology, Skills
Application	**Career Planning** – Students or professionals can use the listed IT roles to identify suitable career paths and plan their future goals. **Skill Development** – The qualities and responsibilities mentioned help individuals understand what skills and ethics they must develop to succeed in IT. **Educational Guidance** – Teachers and learners can use the note to guide course selection, curriculum planning, or choosing IT specializations.
Summary	Information Technology offers various career opportunities such as system analysis, software engineering, AI development, and network management. IT professionals are expected to demonstrate qualities like honesty, confidentiality, discipline, and lifelong learning. Several professional bodies, including NCS, CPN, ACM, and IEEE, support growth and standards in the IT field.
Student engagement: *i.* What have I learnt? *ii*. Skills acquired	
My Question(s)/Evaluation	
Action	

CAREER OPPURTUNITIES IN IT

Career opportunity in I.T refers to any professional occupation that involves the use, application, or development of information technology. These careers are found across all sectors of the economy, and they require specialized knowledge and skills in areas such as programming, networking, data analysis, or hardware management.

Here are some career opportunities and functions in Information Technology (IT):

1. **Computer Manager:** Oversees the overall IT operations, manages IT teams, and ensures IT strategies align with business goals.

2. **System Analyst:** Analyses and optimizes computer systems to meet business needs, identifies problems, and implements solutions.

3. **Computer Educator:** Teaches computer science concepts, trains individuals on software and hardware, and develops educational programs.

4. **Data Analyst/Scientist:** Interprets and analyses data to inform business decisions, identifies trends, and develops predictive models.

5. **Network Administrator:** Designs, implements, and maintains computer networks, ensuring security and efficiency.

6. **Software Engineer:** Designs, develops, tests, and maintains software applications, ensuring they meet user needs.

7. **Computer Engineer and Technician:** Installs, maintains, and repairs computer hardware and software, troubleshoots issues.

8. **Robotics Engineer:** Designs and builds robots and robotic systems.

9. **Database Administrator (DBA):** Installs, configures, and maintains databases to ensure data integrity and security.

10. Cyber security Analyst: Protects computer systems and networks from threats and attacks.

11. **Website Developer/Administrator:** Designs, develops, updates, and maintains websites; is also responsible for website security.

12. **Computer Graphics Artist:** A professional who does complex artworks such as drawing, painting, and animations with the computer

<u>Qualities of good IT professionals</u>:

1. A good Computer professional must have undergone appropriate training either through formal school system or through appropriate professional certificate training programmes.

2. Good communication skill

3. They must be people who pay attention to details.

4. A good computer professional must be trustworthy because he/she will handle sensitive information about the organization.

5. He/she must be knowledgeable and always ready to learn new things.

6. A good computer professional should be able to work with other people because computer based projects often require teamwork.

7. A good IT professional should possess strong math skills are necessary because math is used in many computer applications, such as when dealing with circuits or programming.

8. An Ability to Handle Multitasking. People working with computers are often involved in many tasks at once and must be able to manage all of their responsibilities simultaneously.

9. He must have resting and relaxation time. A good computer must have a resting and relaxation time in order to relax and refresh the brain

10. He must belong to professional body

COMPUTER PROFESSIONAL BODIES

- Some notable computer professional bodies are:
- Nigeria Computer Society (NCS)
- Computer Professionals Registration Council of Nigeria (CPN)
- Association for Computing Machinery (ACM)
- Institute of Management Information System (IMIS)
- Information Technology Association of Nigeria (ITAN)
- Nigeria Internet Group (NIG)
- Nigeria Communication Commission (NCC)
- Internet Service Provider Association of Nigeria (ISPAN)
- IT industry Association of Nigerian (ITIAN)
- Association of Telecom Companies in Nigeria (ATCN)
- Computer Association of Nigeria (COAN)

Date	**WEEK 3**
Topic	**Basic Computer Hardware Components I**
Objectives	By the end of the lesson, students should be able to: i. Identify the internal parts of an open system unit ii. Identify the external parts of system unit iii. Describe how the CPU works with main memory. iv. Identify and explain functions of some external features. v. Connect and power on a complete computer system
Insight	A computer's hardware consists of internal components that handle processing and storage, and external devices that allow users to interact with the system. Data is stored and measured using the binary system, with units ranging from bits to terabytes.
Key words/New terms	Hardware**,** CPU**,** RAM**,** Peripherals**,** Storage Unit
Application	**Understanding Computer Functionality** – Helps learners understand how different hardware components work together to perform input, processing, output, and storage tasks. **Troubleshooting and Maintenance** – Provides knowledge needed to identify hardware parts, diagnose problems, and perform basic computer repairs. **Hardware Selection and Upgrading** – Guides users in choosing suitable components like RAM, storage drives, or peripherals when upgrading or building a computer.
Summary	Computer hardware includes internal components like the CPU, RAM, storage devices, and the motherboard, as well as external peripherals such as the keyboard, mouse, and monitor that allow user interaction. Data is stored and measured using the binary system, with units ranging from bits to terabytes.
Student engagement: *i.* What have I learnt? *ii*. Skills acquired	
My Question(s)/Evaluation	
Action	

Basic Computer Hardware Components I

Computer hardware includes the physical, tangible components of a computer system that

you can see and touch. These components include both the internal parts that perform the core functions, such as the central processing unit (CPU), memory (RAM), and hard drive, and the external devices (peripherals) that allow you to interact with the computer, like the keyboard, mouse, and monitor.

The hardware can be classified into two categories, they are:

1. System unit

2. Peripherals

SYSTEM UNIT

A computer system unit is the case that houses the internal parts of a computer. It's like the "body" of the computer that houses all the important parts that allows the computer to function and perform tasks. The system unit consist of both internal and external components.

Internal components of the system unit

1. **Central Processing Unit (CPU):** The "brain" of the computer that processes instructions. Examples include Core i3, Core i5, Core i7, Core i9, Core Ultra

2. **Random-Access Memory (RAM):** A temporary storage for data that the CPU can access

quickly.

3. **Read only memory (ROM):** ROM (Read-Only Memory) is a type of computer memory that stores permanent data and instructions for the device to start up. It is typically located on the BIOS chip within the motherboard of a personal computer.

4. **Motherboard:** This is the main circuit board where all internal components are connected. It allows communication between the CPU, memory, and other devices.

5. **Power Supply Unit (PSU):** Converts electric power from alternating current (AC) to direct current (DC) for use by the computer components.

6. **Cooling Fan:** Helps to cool down the internal components and prevent overheating.

7. **Hard Disk Drive (HDD):** Stores data and programs permanently in the computer.

8. **Expansion Cards:** These are used to add extra features such as sound, video, or network functions to the computer.

External components of the system unit

- Power button: Turns the computer on or off.
- Reset switch: Restarts the computer.
- Ports: Allow connection of external devices like flash drives, mouse, or keyboard.

- Disk Drives: Used to insert CDs, DVDs, or external storage media.
- Indicator Lights: Show when the system is on or when data is being processed.

PERIPHERALS

A peripheral is an external device that connects to the computer to give it additional functionality, but is not part of the core computer architecture like the CPU or motherboard.

Peripherals fall into three main categories:

Input Devices: Devices used to enter data into the computer. E.g., Keyboard, Mouse, scanner, microphone.

Output Devices: Devices that display or present information from the computer. E.g., Monitor, printer, projector, speaker etc.

Storage Unit: Storage units are hardware that save data and information for current or future use, and they are divided into primary storage (like RAM, ROM) and secondary storage (like Hard Disk Drives and Flash Drives).

Date	**WEEK 4**
Topic	Units of Computer Storage
Objectives	• Define computer storage units. • Identify the different units of storage from bit to yottabyte. • Explain the relationship between storage units. • Convert one storage unit to another correctly.
Insight	Computer storage units are standards used to measure the amount of data a computer can store. Since computers use the binary system, storage units are based on powers of 2. The smallest unit is a bit, while larger units include bytes, kilobytes, megabytes, gigabytes, terabytes, and beyond.
Key words/New terms	Bit, Nibble, Byte, Word, Kilobyte (KB), Megabyte (MB), Gigabyte (GB), Terabyte (TB), Petabyte (PB), Exabyte (EB), Zettabyte (ZB), Yottabyte (YB), Binary System, Data Storage, Conversion.
Application	**Determining File Sizes:** Helps measure the size of documents, images, audio files, videos, and software applications. **Software Development:** Helps programmers manage memory usage and optimize applications for efficient storage. **Computer System Planning:** Helps individuals and organizations choose appropriate storage devices based on their data needs.
Summary	• Computer storage units measure the amount of data stored in a computer. • The smallest unit is the bit (0 or 1), while the byte (8 bits) is the basic storage unit. • Larger units are formed by grouping smaller units (KB, MB, GB, TB, etc.). • To convert from a larger unit to a smaller unit, multiply. • To convert from a smaller unit to a larger unit, divide. • Examples: 1200 bits = 150 bytes, 30 KB = 30,720 bytes, and 10,240 KB = 10 MB.
Student engagement: *i.* What have I learnt? *ii*. Skills acquired	
My Question(s)/Evaluation	
Action	

UNITS OF COMPUTER STORAGE

The unit of storage in computer is the standard of measurement used to measure the size of storage devices. Data can be stored in the computer through several methods using varying electronic components. These methods however are all digital in nature as most computers of today are in digital format. The use of binary system became necessary because of the way the electronic components of the computer work.

Common Units of Storage in the Computer

Here are the fundamental units of digital storage, from the smallest to the largest:

1. Bit**:** Bit is an acronym for Binary digits. It is the absolute smallest unit of data in a computer, representing either a 0 (zero) or a 1 (one).
2. Nibble: A nibble is a collection of four bits.
3. Byte: A byte consists of eight bits. This is a very important unit because it's usually the smallest amount of data that can represent a single character, like a letter or a number. It's considered the fundamental unit of storage that programs work with.
4. Word: A word is the amount of data that a computer's processor (CPU) can handle at one time. Its size can vary depending on the computer, but commonly, it's 16, 32, or 64 bits.
5. Kilobyte (KB): A Kilobyte consists of 2^{10} (1,024) Bytes. While often approximated as 1,000 Bytes, computers use 1,024 because they work in a binary (base-2) system.
6. Megabyte (MB): One Megabyte consists of 2^{20} (1,048,576) Bytes, or 1,024 Kilobytes.
7. Gigabyte (GB): A Gigabyte is a collection of 2^{30} (1,073,741,824) Bytes, or 1,024 Megabytes.
8. Terabyte (TB): A Terabyte consists of 2^{40} (1,099,511,627,776) Bytes, or 1,024 Gigabytes.
9. Petabyte (PB): A Petabyte consists of 2^{50} (1,125,899,906,842,624) Bytes, or 1,024 Terabytes.
10. Exabyte (EB): An Exabyte consists of 2^{60} (1,152,921,504,606,846,976) Bytes, or 1,024 Petabytes.
11. Zettabyte (ZB): A Zettabyte consists of 2^{70} (1,180,591,620,717,411,303,424) Bytes, or 1,024 Exabytes.
12. Yottabyte (YB): A Yottabyte consists of 2^{80} (1,208,925,819,614,629,174,706,176) Bytes, or 1,024 Zettabytes.

CONVERSION FROM ONE UNIT TO ANOTHER

To convert from one unit to another, you only need to remember one simple rule:

- To convert from a larger unit to a smaller unit, you multiply.

- To convert from a smaller unit to a larger unit, you divide.

Example 1: Convert 1200 bits to bytes

Solution:
We need to convert 1200 bits to bytes.
Let the unknown number of bytes be x
So, 1200 bits = x bytes
The relationship between bits and bytes is:
8 bits = 1 byte
Cross-multiplying gives:
8x = 1200
Divide both sides by the coefficient of x
X = 150 bytes

ALTERNATIVELY

Bits are a smaller unit than bytes. So, to convert, we divide.
Since 1 byte = 8 bits, we divide 1200 by 8.
1200 ÷ 8 = 150
Answer: 1200 bits = 150 bytes

Example 2: Convert 30 KB to bytes

Solution: Kilobytes (KB) are a larger unit than bytes. So, to convert, we multiply.
Since 1 KB = 1024 bytes, we multiply 30 by 1024.
30 × 1024 = 30720
Answer: 30 KB = 30,720 bytes

Example 3: Convert 10240 KB to MB

Solution: Kilobytes (KB) are a smaller unit than Megabytes (MB). So, to convert, we divide.
Since 1 MB = 1024 KB, we divide 10240 by 1024.
10240 ÷ 1024 = 10
Answer: 10240 KB = 10 MB

Date	**WEEK 5**
Topic	**Word processing**
Objectives	By the end of the lesson, students should be able to: i. state the uses of word processor ii. state examples of word processor ii. Load and exit word processor ii. Create, save and retrieve files iii. Apply features of word processing as cut, copy and paste.
Insight	A word processor is a software tool that allows users to create, edit, format, and manage text documents efficiently using features like formatting options, tables, and basic file operations.
Key words/New terms	Word Processor, Editing, Formatting, Document
Application	**Document Creation** – Used for writing letters, reports, essays, and other text-based documents. **Office and Administrative Work** – Helps in preparing official files, memos, meeting minutes, and professional documents. **Education and Assignments** – Students and teachers use word processors for assignments, notes, lesson plans, and projects.
Summary	A word processor is a software tool used to create, edit, format, save, and print text documents with features like spell check, tables, and formatting options. It also provides essential functions such as opening, closing, and managing documents, along with tools for typing, editing, and inserting tables.
Student engagement: *i.* What have I learnt? *ii*. Skills acquired	
My Question(s)/Evaluation	
Action	

WORD PROCESSING

Word processing is the use of a computer or digital device to create, edit, format, store, and print text-based documents.

Word Processor

A word processor is a software application that allows users to create, edit, format, store, and print text documents. It goes beyond simple text editing by offering features like spell check, formatting options, and the ability to insert images and tables, making it ideal for a wide range of documents from simple notes to professional reports.

EXAMPLES OF WORD PROCESSORS

- Microsoft Word
- Google Docs
- LibreOffice Write
- WPS Writer
- WordPad
- Notepad
- Corel word perfect
- Word star
- Lotus Notes
- Word perfect
- Perfect Writer
- Word pro
- Professional write

STARTING MICROSOFT WORD

To Start Microsoft Word 2010 from the Start Menu:

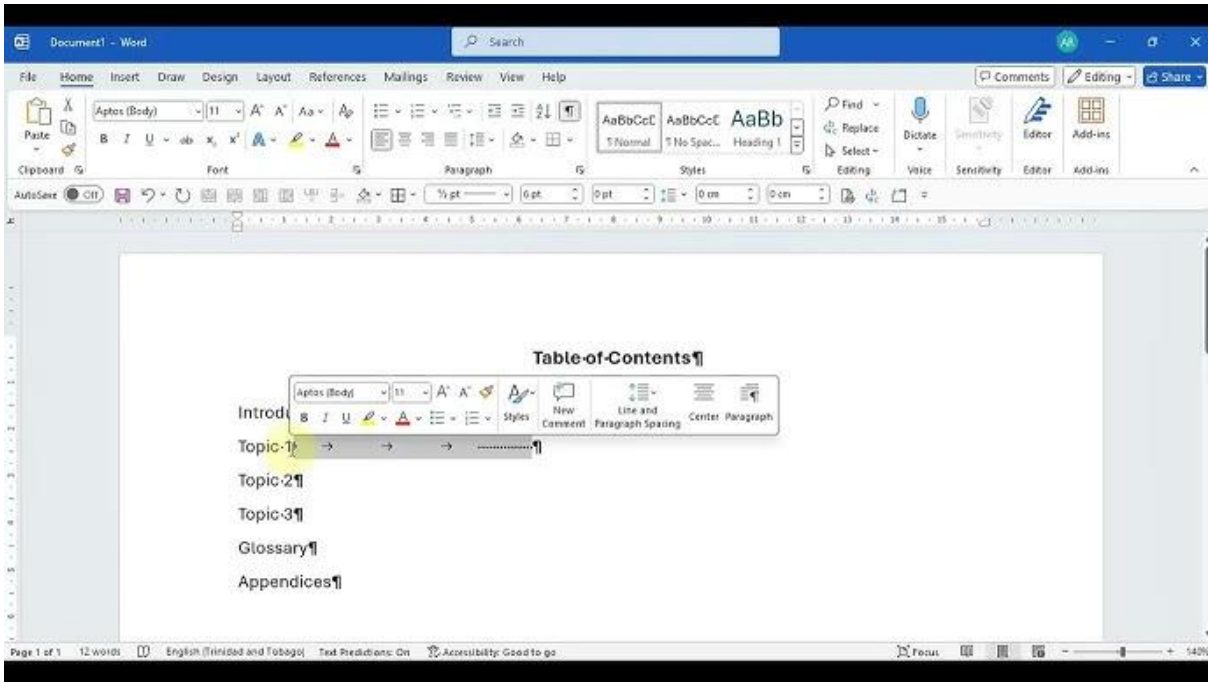
1. Click on the Start button

2. Point to All Programs

3. Click on Microsoft Office

4. Click on Microsoft Word 2010.

This will launch Microsoft Word 2010 application and the first screen that you will see is a new blank page document.



Creating a new document

To create a new document:

- Open MS Word
- Click on blank document
- A blank white page appears where you can type your document

Saving a new document

To save a document:

- Click on File
- Click Save As
- Select the location where you want to save the document
- Type a file name for your document
- Click on save

Closing a new document

To close a document:

- Click on File
- Click Close

Opening a new document

To open a document:

- Click on File
- Click Open
- Select the location where you saved the document
- Click on the document
- Click on open

Inserting Table in MS Word

- Click on Insert menu
- Click on Table
- Select the number of rows and columns
- Use the table tools to edit and format the table

Editing MS Word Documents

Copying and moving text
Word allows you to copy text that's already in your document and paste it in other places, which can save you a lot of time and effort. If you want to move text around in your document, you can cut and paste or drag and drop.

To copy and paste text:
1. Select the text you want to copy.
2. Click the Copy command on the Home tab. You can also press Ctrl on your keyboard.
3. Place the insertion point where you want the text to appear.
4. Click the Paste command on the Home tab. You can also press Ctrl+V on your keyboard.
5. The text will appear.

To cut and paste text:
1. Select the text you want to cut.
2. Click the Cut command on the Home tab. You can also press Ctrl+X on your keyboard.
3. Place your insertion point where you want the text to appear.
4. Click the Paste command on the Home tab. You can also press Ctrl+V on your keyboard.
5. The text will appear.
You can also cut, copy, and paste by right-clicking your document and choosing the desired action from the drop-down menu.

Undo and Redo
Let's say you're working on a document and accidentally deleted some text. Fortunately, you won't have to retype everything you just deleted! Word allows you to undo your most recent action when you make a mistake like this.
To do this, locate and select the Undo command on the Quick Access Toolbar. You can also press Ctrl+Z on your keyboard. You can continue using this command to undo multiple changes in a row.

By contrast, the Redo command allows you to reverse the last undo. You can also access this command by pressing Ctrl+Y on your keyboard.

FORMATTING A WORD DOCUMENT

The process of formatting a document involves specifying how the document will look in its final form on the screen and when printed. Thus, making changes to the appearance of a document. In Word, you have several options for adjusting text, including font, size, and color. You can also adjust the alignment of the text to change how it is displayed on the page.

To change the font size:

1. Select the text you want to modify.
2. On the Home tab, click the Font Size drop-down arrow. Select a font size from the menu. If the font size you need is not available in the menu, you can click the Font Size box and type the desired size, and then press Enter.
3. The font size will change in the document.

You can also use the Grow Font and Shrink Font commands to change the font size.

To change the font:

By default, the font of each new document is set to Calibri. However, Word provides many other fonts you can use to customize text.

1. Select the text you want to modify.
2. On the Home tab, click the drop-down arrow next to the Font box. A menu of font styles will appear.
3. Select the font style you want to use.
4. The font will change in the document.

To change the font color:

1. Select the text you want to modify.
2. On the Home tab, click the Font Color drop-down arrow. The Font Color menu appears.
3. Select the font color you want to use. The font color will change in the document.

Your color choices aren't limited to the drop-down menu that appears. Select More Colors at the bottom of the menu to access the Colors dialog box. Choose the color you want, and then click OK.

To use the Bold, Italic, and Underline commands:

The Bold, Italic, and Underline commands can be used to help draw attention to important words or phrases.

1. Select the text you want to modify.
2. On the Home tab, click the Bold (B), Italic (I), or Underline (U) command in the Font group (depending on the one you wish to use).
3. The selected text will be modified in the document.

PRINTING A DOCUMENT

To access the Print pane:

1. Select the File tab.
2. Select Print. The Print pane will appear.

You can also access the Print pane by pressing Ctrl+P on your keyboard.

3. Navigate to the Print pane, and then select the desired printer.
4. Enter the number of copies you want to print.
5. Select any additional settings if needed.
6. Click Print.

Date	**WEEK 6**
Topic	**Digital Media**
Objectives	• Define digital media. • Identify examples and types of digital media. • Explain the uses of digital media platforms. • Discuss the advantages and disadvantages of digital devices.
Insight	Digital media refers to content such as text, audio, video, and graphics that is created, shared, and accessed using digital technology and electronic devices. It plays an important role in communication, education, business, entertainment, and information sharing.
Key words/New terms	Digital Media, social media, Website, Web Page, E-book, Podcast, Digital Audio, Streaming, MP3, Email Marketing, Blog, Mobile App, Owned Digital Media, Paid Digital Media, Earned Digital Media, E-commerce, Digital Devices.
Application	• Communicating through social media, emails, and messaging apps. • Accessing online learning materials, e-books, and virtual classrooms. • Marketing products and services through digital advertising. • Streaming music, videos, and podcasts for entertainment. • Buying and selling goods online through e-commerce platforms. • Networking and collaborating with people and organizations worldwide.
Summary	• Digital media includes content created and shared using digital technology. • Examples include social media, websites, e-books, podcasts, digital audio, blogs, email marketing, and mobile apps. • The three main types of digital media are Owned Media, Paid Media, and Earned Media. • Digital media platforms are used for communication, education, business, entertainment, information sharing, e-commerce, and networking. • Digital devices offer benefits such as fast communication, easy access to information, productivity, and convenience. • However, they also have disadvantages such as health issues, addiction, privacy risks, high costs, and electronic waste.
Student engagement: *i.* What have I learnt? *ii*. Skills acquired	
My Question(s)/Evaluation .	
Action	

Digital Media

What is Digital Media?

Digital Media refers to any content such as text, audio, video, and graphics that is created, shared, and accessed using electronic devices and digital technology.

Examples of digital media platforms

- Social media
- Websites and web pages
- Electronic books and podcasts
- Digital audio such as streaming and mp3s
- Email marketing and blogs
- Mobile apps

Social Media

Social media platforms allow people and organizations to create, share, and interact with content online. They are used for communication, marketing, entertainment, and networking.

Examples: Facebook, WhatsApp, Instagram, X, TikTok etc.

2. Websites and Web Pages

A website is a collection of related web pages accessible through the internet. Web pages provide information, services, or entertainment to users.

Examples: An online store, A university website, A news website, A company homepage

3. Electronic Books (E-books) and Podcasts

E-books are digital versions of printed books that can be read on computers, tablets, e-readers, or smartphones.

Podcasts are digital audio programs that users can stream or download and listen to on demand.

4. Digital Audio (Streaming and MP3s)

Digital audio refers to sound content stored and distributed electronically. Users can either stream it online or download audio files such as MP3s.

5. Email Marketing and Blogs

Email marketing involves sending promotional or informational messages directly to subscribers' email inboxes.

Blogs are regularly updated websites or sections of websites containing articles, opinions, tutorials, or news.

6. Mobile Apps

Mobile apps are software applications designed to run on smartphones and tablets. They provide services, entertainment, communication, shopping, and productivity tools.

Examples: Banking apps, Shopping apps, educational apps, Fitness apps

Types of Digital Media

There are three major types of digital media, namely:

1. **Owned Digital Media:** Owned digital media is managed entirely by the organization, brand, or individual that owns it. This allows them to decide what message, design, and content they want to get out. Examples include:
    - Companies have complete control over the functionality and content of their websites, blogs, and any other digital content.
    - Brands develop and manage mobile apps so they can directly interact with customers.
    - Businesses create official accounts on social media sites and control any content they post.

2. **Paid Digital Media:** When a business pays to advertise content on platforms like websites, search engines, or social media apps, this is paid digital media. Examples include Google Ads, display ads, YouTube ads, and social media ads.
3. **Earned Digital Media:** Think of it as free word-of-mouth publicity online. Earned digital media is something that brands and companies can be especially proud of. This is exposure gained through customer reviews, media coverage, and social media mentions.

Uses of Digital media Platforms

1. **Communication**: Digital media platforms allow people to send messages, make calls, and hold video conferences. Example: social media, email, and messaging apps.
2. **Education and Learning**: They provide access to online courses, e-books, tutorials, and virtual classrooms. They enable distance learning and skill development.
3. **Business and Marketing**: Digital media platforms help businesses promote products and services to a wider audience. They support online advertising, customer engagement, and sales.
4. **Entertainment**: They offer movies, music, games, podcasts, and other digital content. They allow users to stream or download media.
5. **Information Sharing**: Digital media platforms enable quick access to news, research materials, and educational resources. Allow users to create and share content.
6. **E-commerce**: Facilitate buying and selling of goods and services online.
7. **Networking and Collaboration**: they provide means for professionals, students, and organizations to connect and share ideas

Advantages and Disadvantages of Digital Devices

Digital devices are electronic tools that process, store, and transmit digital information, such as smartphones, computers, tablets, and smartwatches.

Advantages of Digital Devices

1. **Fast Communication**: Enable instant messaging, emails, and video calls.
2. **Easy Access to Information**: Provide quick access to educational materials, news, and online resources.
3. **Increased Productivity**: Help users' complete tasks efficiently through software and applications.
4. **Convenience**: Allow users to perform many activities, such as banking, shopping, and learning, from anywhere.
5. **Data Storage**: Store large amounts of information in a small space.
6. **Entertainment**: Provide access to games, music, movies, and social media.
7. **Improved Learning**: Support online education, research, and interactive learning experiences.

Disadvantages of Digital Devices

1. **Health Issues**: Excessive use can cause eye strain, poor posture, and sleep problems.
2. **Addiction and Distraction**: Users may spend too much time on social media, games, or entertainment.
3. **Privacy and Security Risks**: Personal information may be exposed to hackers, scams, or cybercrime.
4. **Cost**: Purchasing and maintaining devices can be expensive.
5. **Dependence on Technology**: People may become overly reliant on devices for everyday tasks.
6. **Reduced Face-to-Face Interaction**: Excessive device use can decrease direct social interaction.
7. **Electronic Waste**: Discarded devices contribute to environmental pollution.

Date	**Week 7**
Topic	**Safety and Security Features on Digital Devices**
Objectives	**Students should be able to:** 1. identify safety and security features on digital devices 2. explain the usefulness of GPS as safety measures on digital devices 3. list ways of securing digital devices 4. state importance of securing digital devices
Insight	Safety and security features are tools and settings that help protect digital devices, personal information, and users from unauthorized access, malware, and online threats.
Key words/New terms	1. **Safety** - Features like GPS, anti-glare screens, and parental controls that protect users physically and help manage device use. 2. **Security** - Tools like passwords, 2FA, and security keys that verify identity and block unauthorized access. 3. **GPS** - A satellite-based system used for locating devices, navigation, emergency assistance, and safety monitoring.
Application	1. **Personal Device Protection** Using screen locks, passwords, fingerprint/retina locks, and anti-virus to keep phones, tablets, and laptops safe from unauthorized access, theft, and malware. 2. **Child Online Safety** Parents apply parental controls and GPS monitoring to manage what kids access online and track their location for safety during school, trips, or daily activities. 3. **Emergency Response & Recovery** *GPS* helps locate lost or stolen devices and shares exact location with emergency responders so they can reach someone quickly during accidents or danger.
Summary	**Safety and security features** are tools that protect devices, personal info, and users from threats. **Safety features** focus on physical/user protection: • **Anti-glare screen** reduces light glare • **GPS** finds device location, aids navigation, emergencies, and monitoring • **Parental controls** let guardians manage kids' device use
Student engagement: i. What have I learnt? ii. Skills acquired	
My Question(s)/Evaluation	
Action	

Safety and Security Features on Digital Devices

Safety and security features are tools and settings that help protect digital devices, personal information, and users from unauthorized access, malware, and online threats.

A. <u>Safety features</u>:

- Anti-Glare Screen: An anti-glare screen is a clear panel or filter placed over a screen that helps prevent sun and light glare on a screen.

- Global Positioning System (GPS): GPS is a satellite-based navigation system that helps determine the exact location of a device anywhere on Earth.

- Parental Controls: Parental controls are settings that allow parents or guardians to manage how children use digital devices and online services.

B. <u>Security features</u>:

- Security key: A security key is a physical device used to verify a user's identity when logging into an account.

- Anti-Virus: An anti-virus program is software designed to detect, prevent, and remove malicious software (malware) from a device.

- Password: A password is a secret combination of letters, numbers, and symbols used to verify a user's identity

- Finger prints lock: A fingerprint lock uses a person's unique fingerprint to unlock a device.

- Retina lock: A retina lock (or iris/eye scan lock) uses the unique patterns in a person's eye to verify identity.

- pattern lock: A pattern lock requires the user to draw a specific pattern on the screen to unlock the device.

- Two-Factor Authentication (2FA): Two-Factor Authentication (2FA) is a security method that requires two forms of verification before access is granted.
-
- Screen Lock: A screen lock is a security feature that automatically or manually locks a device when it is not in use.

<u>Usefulness of GPS as a Safety Measure on Digital Devices</u>

Its usefulness includes:

1. Locating Lost or Stolen Devices: GPS helps users find their phones, tablets, or laptops if they are lost or stolen through device-tracking services.

2. Emergency Assistance: During emergencies, GPS can provide a user's exact location to emergency responders, helping them reach the person quickly.
3. Navigation and Direction: GPS guides users to their destinations safely by providing accurate maps, routes, and traffic information.
4. Safety Monitoring: Parents and guardians can use GPS-enabled apps to know the location of children and ensure their safety.
5. Travel Safety: GPS helps travellers avoid getting lost by showing their current location and suggesting the best routes.
6. Location Sharing: Users can share their location with trusted family members or friends, which can be useful during trips, outdoor activities, or emergencies.

Importance of Securing Digital Devices

Securing digital devices such as smartphones, tablets, and computers is important for the following reasons:

1. Protects Personal Information: Prevents unauthorized people from accessing personal data such as photos, messages, contacts, and documents.
2. Prevents Identity Theft: Helps protect sensitive information that could be used by criminals to impersonate the user.
3. Protects Financial Information: Keeps banking details, credit card information, and online payment accounts safe from cybercriminals.
4. Protects Against Malware and Viruses: Security measures help prevent malicious software from damaging the device or stealing information.
5. Reduces the Risk of Cyberattacks: Strong security features help defend against hacking, phishing, and other online threats.

Date	**WEEK 8**
Topic	**E-Waste (Electronic Waste)**
Objectives	**Students should be able to:** 1. State the meaning of e-waste. 2. Examples of e-waste (scrap computers, phones, batteries, etc). 3. Harmful substances found in e-waste
Insight	E-waste refers to discarded, obsolete, broken, or unwanted electronic devices and electrical equipment that are no longer in use.
Key words/New terms	E-waste, Hazardous, Pollution, Recycle
Application	1. **Environmental Protection:** Managing e-waste properly to prevent soil, water, and air pollution from hazardous substances like lead, mercury, and PVC. Helps protect ecosystems and biodiversity. 2. **Public Health & Safety:** Understanding how toxic materials like cadmium, arsenic, and beryllium cause kidney damage, cancer, and lung diseases. Guides safe handling to protect workers and communities. 3. **Recycling & Resource Recovery:** Recycling e-waste through approved centres to recover valuable materials like gold, copper, and aluminium instead of wasting natural resources.
Summary	**E-waste** means discarded, obsolete, or broken electronic devices like phones, laptops, TVs, batteries, and cables. **Harmful substances** in e-waste include lead, mercury, cadmium, arsenic, beryllium, and lithium. These can cause brain damage, kidney problems, cancer, lung disease, and fires if not handled properly.
Student engagement: iii. What have I learnt? iv. Skills acquired	
My Question(s)/Evaluation	
Action	

E-Waste (Electronic Waste)

E-waste refers to discarded, obsolete, broken, or unwanted electronic devices and electrical equipment that are no longer in use.

Examples of e-waste include:

- Mobile phones
- Computers and laptops
- Tablets
- Televisions
- Printers
- Radios
- Batteries
- Chargers and cables

Harmful Substances Found in E-Waste

E-waste contains several hazardous substances that can harm human health and the environment if not handled properly. These include:

1. **Lead**: Found in old computer monitors, televisions, and circuit boards. It can damage the brain, nervous system, and kidneys.
2. **Mercury**: Found in LCD screens, switches, and some batteries. Can affect the brain, nervous system, and unborn babies.
3. **Cadmium**: Found in rechargeable batteries and semiconductors. Can cause kidney damage and weaken bones.
4. **Chromium (Hexavalent Chromium)**: Used in metal coatings and electronic components. Can cause skin irritation and increase cancer risk.
5. **Arsenic**: Found in some electronic components and semiconductors. Can cause skin disorders, poisoning, and cancer.
6. **Beryllium**: Found in connectors, motherboards, and circuit boards. Can cause lung diseases when inhaled.
7. **Brominated Flame Retardants (BFRs)**: Used in plastic casings, cables, and circuit boards. Can affect the hormone and nervous systems.
8. **Polyvinyl Chloride (PVC)**: Found in cable insulation and plastic parts. Releases toxic gases when burned.
9. **Nickel**: Found in batteries and electronic components. Can cause allergic reactions and respiratory problems.

10. **Lithium**: Found in lithium-ion batteries used in phones and laptops. Can cause fires and environmental contamination if improperly disposed of.

Environmental Impacts of E-Waste

1. **Soil Pollution** – Toxic chemicals from e-waste can leak into the soil, making it less fertile and harmful to plants.
2. **Water Pollution** – Hazardous substances can contaminate rivers, lakes, and groundwater, affecting aquatic life and human water supplies.
3. **Air Pollution** – Burning e-waste releases poisonous gases and smoke into the atmosphere, reducing air quality.
4. **Harm to Wildlife** – Animals can be poisoned by contaminated soil, water, or food, leading to illness or death.
5. **Loss of natural Resources** – Valuable materials such as gold, copper, and aluminium are wasted when e-waste is not recycled.
6. **Increased greenhouse gas Emissions** – More energy is needed to produce new electronic devices when old ones are not recycled, contributing to climate change.
7. **Damage to ecosystems** and **biodiversity**– Toxic substances from e-waste can disrupt natural habitats and affect the balance of ecosystems.

Proper Management of E-Waste

- Recycle electronic devices through approved recycling centres.
- Repair and reuse devices when possible.
- Donate working electronics to others.
- Dispose of old electronics through authorized e-waste collection programs.

Date	**WEEK 9**
Topic	**Introduction to Graphic Design and Canva**
Objectives	Students should be able to: 1. explain the meaning of graphic design and its importance; 2. apply the principles of balance, contrast, and alignment in a simple graphic design project; 3. identify key tools in Canva; 4. use Canva templates to create designs
Insight	Graphic design is the creative process of combining text, symbols, and images to communicate a specific message or concept visually.
Key words/New terms	Graphic design, Canva, Templates, Elements, Brand Kit
Application	1. **Social Media Marketing** Creating posts, videos, stories, and carousels for Instagram, Facebook, Pinterest using Canva templates, text tools, and the content planner to schedule directly to platforms. 2. **Business Branding & Promotion** Designing logos, flyers, posters, and brochures with the Brand Kit to keep colours, fonts, and logos consistent. Used for advertising products/services and brand recognition. 3. **Presentations & Documents** Building slides with presentation tools like presenter view, recording, and animations. Also exporting PDFs for pricing guides, resumes, and business reports.
Summary	Graphic design is the creative process of combining text, symbols, and images to communicate visually. A graphics package is software used to create and edit images — examples include Corel Draw, Photoshop, Illustrator, and Canva.
Student engagement: v. What have I learnt? vi. Skills acquired	
My Question(s)/Evaluation	
Action	

Introduction to Graphic Design and Canva

Graphic design is the creative process of combining text, symbols, and images to communicate a specific message or concept visually.

A graphics package is an application that can be used to create and manipulate images on a computer.

Examples of graphic design include:

- Logos
- Posters
- Flyers
- Brochures

Examples of graphics packages

1. Corel draw
2. Adobe photoshop
3. Canva
4. Paintshop pro
5. Adobe illustrator
6. Harvard graphics
7. Print artist

Introduction to Canva

Canva is a free-to-use online graphic design tool. Use it to create social media posts, presentations, posters, videos, logos and more.

Canva tools

The key tools in Canva can be grouped into a few core categories:

1. Design Editor: **This is the central workspace where you:**

- Drag and drop elements
- Add text, images, videos, and graphics
- Resize and arrange objects
- Apply colours, fonts, and effects

2. Templates

- Ready-made designs for presentations, social media posts, flyers, resumes, logos, and more
- Customizable layouts that speed up design creation

3. Elements Library

Contains:

- Shapes
- Icons
- Illustrations
- Stickers
- Charts
- Frames and grids
- Lines and design components

4. Photo Editing Tools

- Crop and resize
- Filters and adjustments
- Background Remover
- Magic Eraser
- Image enhancement tools

5. Video Editing Tools

- Timeline editing
- Trimming and splitting clips
- Transitions
- Audio tracks and sound effects
- Video animations

6. Text Tools

- Font selection
- Text effects (shadow, curve, outline, etc.)
- Text animations
- Typography templates

7. AI Tools (Magic Studio)

Includes:

- Magic Design
- Magic Write
- Text to Image
- Magic Edit
- Magic Eraser
- Magic Expand
- Magic Animate

8. Brand Kit

- Store logos
- Save brand colours
- Manage fonts
- Maintain consistent branding across projects

9. Collaboration Tools

- Real-time co-editing
- Comments and feedback
- Team folders
- Approval workflows

10. Presentation Tools

- Presenter View
- Presentation recording
- Presenter notes
- Audience engagement features

11. Content Planner

- Schedule social media posts
- Manage publishing calendars
- Post directly to supported platforms

12. Export & Publishing Tools

- Download as PNG, JPG, PDF, MP4, GIF, and other formats
- Share via links
- Print products through Canva Print
- Publish websites

Step 1: Editing Text with Canva's Text Tools

Text is a key component in any design, whether it's a logo, a pricing guide, or a social media post. Canva's text tools allow you to adjust fonts, size, spacing, and effects with ease.

How to Edit Text in Canva:

- Click on the text box you want to edit and type in your content.
- Use the font dropdown in the top toolbar to choose from thousands of fonts.
- Adjust size, spacing, and alignment for readability.
- Apply text effects like shadow, outline, or curve for a unique look.

Step 2: Customising Colours & Using Brand Kits

One of the quickest ways to make a template match your brand is by updating the colours. If you have brand colours, applying them consistently across all designs helps with brand recognition and professionalism.

How to Change Colours in Canva:

- Click on any element – text, shapes, or backgrounds (Step 1 in image below)
- Use the colour picker in the toolbar to choose a new colour

Step 3: Uploading & Replacing Images in Canva

Canva makes it super easy to replace placeholder images with your own photos. This is essential for making templates personalised and unique.

How to Replace Images in Canva:

- Click on an image you want to replace.
- Click ‘Uploads’ on the left panel and upload your own images.
- Drag and drop your image into the placeholder—it will automatically resize.

Step 4: Using Grids, Layers & Alignment Tools for Perfect Layouts

Proper alignment is essential for professional-looking designs. Canva’s grids, layers, and alignment tools help you structure layouts effectively.

How to Use Grids in Canva:

- Click ‘Elements’ > ‘Grids’ to structure your layout easily.
- Choose from single or multi-photo grids for carousels or collages.
- Drag and drop images into grid sections for auto-cropping.

How to Use Layers & Alignment Tools:

Layers: Move objects forward or backward to arrange overlapping elements.
Position Tool: Align text and images precisely within your design.
Group Tool: Select multiple elements and group them together for easier movement and resizing.

Step 5: Saving, Downloading & Sharing Your Designs

Once your template is fully customised, it’s time to download and use it!

Best File Formats for Exporting:
PNG – Best for Instagram, Facebook & Pinterest posts.
JPEG – Smaller file size, good for website uploads (included on Canva free version)
MP4/GIF – For animated graphics.
PDF – Best for pricing guides, brochures & print documents.

How to Download in Canva:
Click “Share” in the top-right corner.
Select “Download”, choose your file format, and save it to your device.